

Outcome and Impact

- Backend engineer experienced in designing, building, and scaling production backend systems across enterprise and early-stage startup environments. Delivered high-impact services handling **APIs, data pipelines, asynchronous workloads, and cloud deployment**, contributing to systems backed by **\$100K+** in venture and research funding.
- **Reduced API latency by 50%** and cut scraping failures from **40% to 5% (Scrapy)** by refactoring monoliths into event-driven microservices.
- **Improved API** query throughput **by 70%** by designing and deploying optimized backend search services **using Django and managed vector stores**.
- Scaled backend **batch processing pipelines 20X** and reduced analytics latency by **50%** using **containerized, cron-driven** job orchestration.
- Built and optimized a **real-time backend service** that reduced **packet latency by 30%** through low-latency networking and event-driven processing.
- Partnered with **product stakeholders** and **customers** to validate backend requirements and **derisk system design decisions**.

Technologies and Languages

- **Web & Systems:** Python, C#, React, .NET, Typescript, FastAPI, Django, Flask, REST APIs, Event-Driven Architecture, Distributed Systems, Redis, PostgreSQL, MongoDB, DynamoDB, Scrapy.
- **Devops and Cloud:** Docker, Azure, AWS (ECS, Lambda, S3, EC2, Databases, SNS), GitHub Actions, CI/CD, CloudWatch.
- **Data Processing:** Batch processing, ingestion pipelines, API orchestration, Redis Queue, Kafka.
- **Testing & Observability** Postman, Selenium, Puppeteer, Sentry.
- **Authentication:** OAUTH, PASETO, JWT.

Work Experience

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| <b>Software Engineer - Backend Systems</b><br>Communications Team | <b>TELUS Communications</b><br>Calgary, Canada | <b>May 23 -Aug 23 and Jan 24 -Aug 24</b> |
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Demo

- Increased backend pipeline test throughput by **50%** by automating **CI/CD for real-time services**.
- Reduced real-time service packet latency by **30%** through backend **performance optimization and low-latency messaging using pub/sub**.
- Delivered a production-ready backend service **deployed on Linux**, supporting high-throughput real-time workloads.
- Built real-time backend processing pipelines for high-volume media data (OpenCV, ffmpeg) ingestion and transformation.
- Implemented **event-driven backend systems** enabling scalable, low-latency computation.
- Integrated monitoring, logging, and external service integrations to improve system reliability and observability.
- Contributed to biweekly stakeholder sessions for scoping, prioritization, and technical design.

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| <b>Founding Backend Engineer</b><br>Product team | <b>ForyourResearch</b><br>Calgary, Canada | <b>Jan 2024 - May 2025</b> |
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Demo

- Worked with stakeholders to define backend requirements and system scalability constraints.
- Increased semantic search throughput by **70%** with optimized vector search pipelines.
- Reduced scraping/API errors **40% to 5%** by moving to event-driven microservices (Redis).
- Containerized and deployed all services to **AWS (ECS, Lambda)**, enabling scalable multi-tenant inference.
- Architected microservices using **Django, Redis queues, AWS ECS/Lambda, Kafka**.

- Built a backend summarization service with **async job handling and external API orchestration using Django**.
- Designed and maintained external **API integrations with observability** + retry logic.

## Backend- Data Engineer

Twimbit

Aug 2021 - Dec 2021

Data Science Team

Singapore, Remote

- Scaled ingestion throughput **20x** using Docker + cron-based pipeline automation.
- Reduced analytics refresh latency **50%** by building KPI microservices.
- Improved clustering and recommendation relevance with semantic similarity APIs.
- Automated API testing with Postman and Sentry, increasing release confidence.

## Full Stack Engineer

Nagarro

Feb 2021 - June 2021

Internal Team

Germany, Remote

- Developed a internal tool using **React and . NET** to find relevant internal company post.
- Developed a writeup application using **HTML, CSS, Javascript, and Node.js** and implemented REST APIs to communicate with internal APIs.

## Backend Developer

Pawitraa

May 2019- Jan 2021

Research and Integration Team

India, South Korea

- **Built and integrated core technical architecture**, supporting backend systems and ML-driven research workflows.
- **Applied data mining and analytical decision-making** across interdisciplinary backend, data, and research problems.
- **Led and mentored 6 technical interns**, delivering research outputs under operational and technical constraints.

## Deployed Projects

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- **ForYourResearch:** ForYourResearch is an open-source platform that automates systematic literature reviews and research discovery using semantic search built using Django, React, AWS, Docker, Langchain, OpenAI ([link](#)).
- **TELUS Remote Support System:** Developed a real-time XR Digital Twin SDK in Unity and AWS that reduced scan-to-sync latency by 30% and improved 3D LiDAR accuracy for seamless remote support and visualization. ([link](#))
- **AI for Energy Prediction in Households:** The system proposes a hybrid CNN-GRU deep learning model that predicts smart home energy consumption more accurately than traditional ML models, improving daily prediction MAE by 17.4% over LSTM. ([link](#))
- **Tree-Based Classification of Firewall Logs to Detect Intrusions:** Developed a machine learning-based intrusion detection system that analyzes firewall log data to identify and prevent malicious network activity. Compared multiple ML algorithms to classify incoming network traffic and actions for firewall enforcement. ([link](#))
- **Artificial Intelligence for Low-Latency and Predictive Intrusion Detection in Power Systems:** Developed an AI-driven intrusion detection framework to secure real-time power system networks against sophisticated cyber threats. Designed a benchmark model using ensemble machine learning techniques for high-accuracy, low-latency threat identification. ([link](#))
- **A Machine Learning Framework for Security and Privacy:** The system introduces a Binomial-Fuzzy Logic integrated with XGBoost-based predictive modelling to identify trusted relationships & mitigate privacy threats in social networks, achieving 92.03% accuracy in friend-foe classification. ([link](#))

## Education

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**University of Calgary, Alberta. 2023**

*Masters in Computer Science, 4.0/4.0*